

Arkadeep Das

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RESEARCH INTERESTS

Cybersecurity, particularly software and network security, cloud security, and the security of LLM-based systems.

EDUCATION

BRAC University, Dhaka, Bangladesh

Jan 2019 | Dec 2023

Bachelor of Science in Computer Science and Engineering

CGPA: 3.55/4.00 (with Distinction, Dean's list for three semester)

Thesis Title: Predicting Novel Coronavirus (nCoV) strains detecting the mutation process applying neural networking.

Relevant Coursework: Cryptography and Cryptoanalysis (GPA:4.0/4.0), Blockchain and Cryptography (GPA:3.7/4.0)

PUBLICATIONS

Conference paper

Published

- AHS Rafi, **A Das**, MT Reza. Improving Performance of Vision Transformer on Skin Lesion Detection Leveraging Domain Associated Transfer Learning. *International Conference on Advances in Modern Age Technologies for Health and Engineering Science (AMATHE)*. DOI: 10.1109/AMATHE61652.2024.10582204
- AHS Rafi, **A Das**, M Das, A Shakil. Forecasting Mutated Genome Sequences of Human Coronaviruses by Unveiling Genomic Mutations Leveraging the Seq2Seq Approach. *2024 IEEE International Conference on Machine Learning and Applied Network Technologies (ICMLANT)*. DOI: 10.1109/ICMLANT63295.2024.00019

RESEARCH EXPERIENCE

Elite Research Lab

Research Assistant

Jun 2026 | Present

- Currently engaging in literature review, research discussions, and early-stage problem exploration to develop future research directions in cybersecurity.

BRAC University

Undergraduate Researcher

Dhaka, Bangladesh

Jan 2023 | Mar 2024

- Explored multiple deep learning RNN architectures, including LSTM and GRU models for forecasting human coronavirus genome mutations, and developed an encoder-decoder Seq2Seq model as the core methodology for my undergraduate thesis.
- Designed a chronological sequence-based approach to learn mutation patterns from previous coronavirus genome data.
- Implemented the full research workflow, including data preprocessing, model development, training, validation, and result analysis.
- Extended the thesis work into a peer-reviewed publication by refining the methodology, experiments, and manuscript.
- Contributed to a separate medical image analysis study that applied domain-associated transfer learning to enhance Vision Transformer performance in skin lesion detection.
- Co-authored two peer-reviewed publications, gaining experience in independent thesis research, collaborative research, experimentation, and academic writing.

INDUSTRY EXPERIENCE

Samsung R&D Institute Bangladesh, SRBD

System Developer Intern

Dhaka, Bangladesh

Nov 2024 | Sep 2025

- Developed enterprise web systems using Laravel, CodeIgniter, PHP, MySQL, JavaScript, and Bootstrap.
- Built attendance, leave, and reporting automation modules, reducing manual workload.
- Designed database-driven workflows for multiple projects.
- Fixed multiple logical and database-related bugs by improving application logic, queries, and system structure.
- Implemented access restriction for an internal voting system in a controlled office environment, gaining exposure to endpoint-based access control concepts.

Fiber@Home Ltd.

IP Transmission Network Intern

Dhaka, Bangladesh

Mar 2024 | Nov 2024

- Worked with routing, switching, network troubleshooting, and traffic analysis in a real-world network environment.
- Gained hands-on experience with Cisco routers/switches and network monitoring tools.
- Assisted in diagnosing connectivity, configuration, and performance-related network issues.
- Developed practical understanding of network infrastructure, which supports future research interests in network security.

SELECTED PROJECTS

Votex: Secure multi-group voting platform

2025

Samsung R&D Institute Bangladesh, SRBD

- Developed an internal voting system with group-wise score tracking, automated result aggregation, and **MAC-based access control** to ensure usage only from authorized devices in a controlled environment.
- Applied bell-curve-based score adjustment to reduce bias and practical solutions to endpoint-based access control and software security considerations.

ProjectMetric: Corporate project performance evaluator

2025

Samsung R&D Institute Bangladesh, SRBD

- Developed a **multi-level admin system** for project scoring, evaluation, and role-based input management.
- Designed dynamic workflows to support separate user roles and automated evaluation logic.

CoreSync: Centralized enterprise operations system

2024 | 2025

Samsung R&D Institute Bangladesh, SRBD

- **Automated** attendance, leave, weekly reporting, award dashboard, project tracker and DOI management features to support HR and administrative workflows.
- Resolved **30+ backend, frontend and logical** issues, improving system functionality and user experience.
- **Optimized database** structures and queries to improve data handling and overall system performance.

AWARDS, HONORS & ACHIEVEMENTS

Dean's List, BRAC University

Awarded for maintaining a CGPA above 3.75 for three semesters.

Recognition of Excellence, Samsung RD Institute Bangladesh, SRBD

Received Recognition of Excellence (Q3 2024) with the development team for outstanding performance and contribution to organizational goals.

LEADERSHIP ACTIVITIES

Instructor

Samsung R&D Institute Bangladesh, SRBD

2025

- Served as an Instructor for SRBD's **Junior Software Academy**, teaching underprivileged students C programming, Microsoft Office tools, and graphic design as part of a CSR-based digital literacy and STEM outreach program.

SKILLS

- **Programming:** Java, Python, PHP, JavaScript, TypeScript
- **Development:** Laravel, Codeigniter, React, Next.js, MySQL, Git
- **AI/ML:** Transfer Learning, Vision Transformers, Seq2Seq Modeling, Medical Image Analysis, Bioinformatics Data Analysis, Model Training & Evaluation
- **Cybersecurity Tools & Skills:** Kali Linux, Wireshark, Nmap, Burp Suite, Network Scanning, Packet Analysis, Vulnerability Analysis, Linux Security, Network Troubleshooting

REFERENCES

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